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New cyathane diterpenoids with neurotrophic and anti-neuroinflammatory activity from the bird's nest fungus *Cyathus africanus*

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ABSTRACT

Eleven new cyathane diterpenoids, designated cyafricanins A–K (**1** – **11**), were isolated from the culture broth of the baisidiomycete *Cyathus africanus* (Nidulariaceae, Bird's nest fungi). Their structures were elucidated by comprehensive analysis of their NMR and HRESIMS data. Cyafricanins A (**1**) was found to possess an unusual 3,4-*seco*-carbon skeleton. All compounds were evaluated for their neurotrophic activity in PC-12 cells and anti-neuroinflammatory activity in BV2 microglia cells. All of the diterpenoids showed nerve growth factor induced neurite outgrowth-promoting activity at concentration of 20 μM. Among them, cyafricanin B (**2**) and cyafricanin G (**7**) exhibited promising neurotrophic activity, and cyafricanin A (**1**) showed strong inhibitory effects on both inducible nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2) expression. Furthermore, molecular docking studies revealed that cyafricanin A (**1**) showed strong interactions with the iNOs protein in the active cavity.

1. Introduction

The genus *Cyathus* belongs to the family Nidulariaceae (Basidiomycota), which are also referred to in the literature as "Bird's Nest Fungi" because their basidiomes resemble the nests of birds and the peridioles that contain the basidiospores are reminiscent of eggs. The characteristic secondary metabolites of the mycelial cultures of *Cyathus* species and other Nidulariaceae are cyathane diterpenoids, which have been isolated first from these Basidiomycota during the 1970s and belong to the fungi-specific classes of terpenoids that do not apparently occur in plants [1]. Cyathane diterpenoids typically possess an unique 5/6/7 tricyclic skeleton, but exceptionally this characteristic ring system may be modified; hence some cyathane derivatives were found to possess a 3,4-*seco* or a 4,9-*seco*-carbon skeleton. Many of these diterpenoids have actually been found during the course of bioprospecting programs, where they exhibited anti-microbial and anti-proliferative activities. Most importantly, some cyathanes were found to possess neurotrophic properties, owing to the fact that they can stimulating the synthesis of neurotrophic proteins, such as Nerve Growth Factor (NGF) and Brain Derived Nerve Growth Factor (BDNF) [1–6]. Our previous research found that 3,4-*seco*-cyathane diterpenoids also have anti-neuroinflammatory activities that are exerted by inhibiting the expression of certain pro-inflammatory enzymes [7–10]. Since it has been well established that neuroinflammation is implicated in the pathogenesis of various neurodegenerative diseases including Alzheimer's disease (AD) [11–13], cyathanes represent promising lead structures for the development of neuroprotective agents.

Cyathus africanus produced quite a different range of cyathanes depending on the fermentation conditions and the duration of the fermentation. In our previous studies, eighteen new cyathane diterpenoids were isolated from different cultures of *C. africanus*. In this study, *C. africanus* was cultured on cooked rice for six months, much longer than normal fermentation time. Chemical investigation led to the isolation of eleven new cyathane diterpenoids, for which we propose the trivial names, cyafricanins A–K (**1–11**) (Fig. 1). Here, we report the isolation, structure elucidation, bioactivities related to neurodegenerative diseases and molecular docking of the bioactive compounds.

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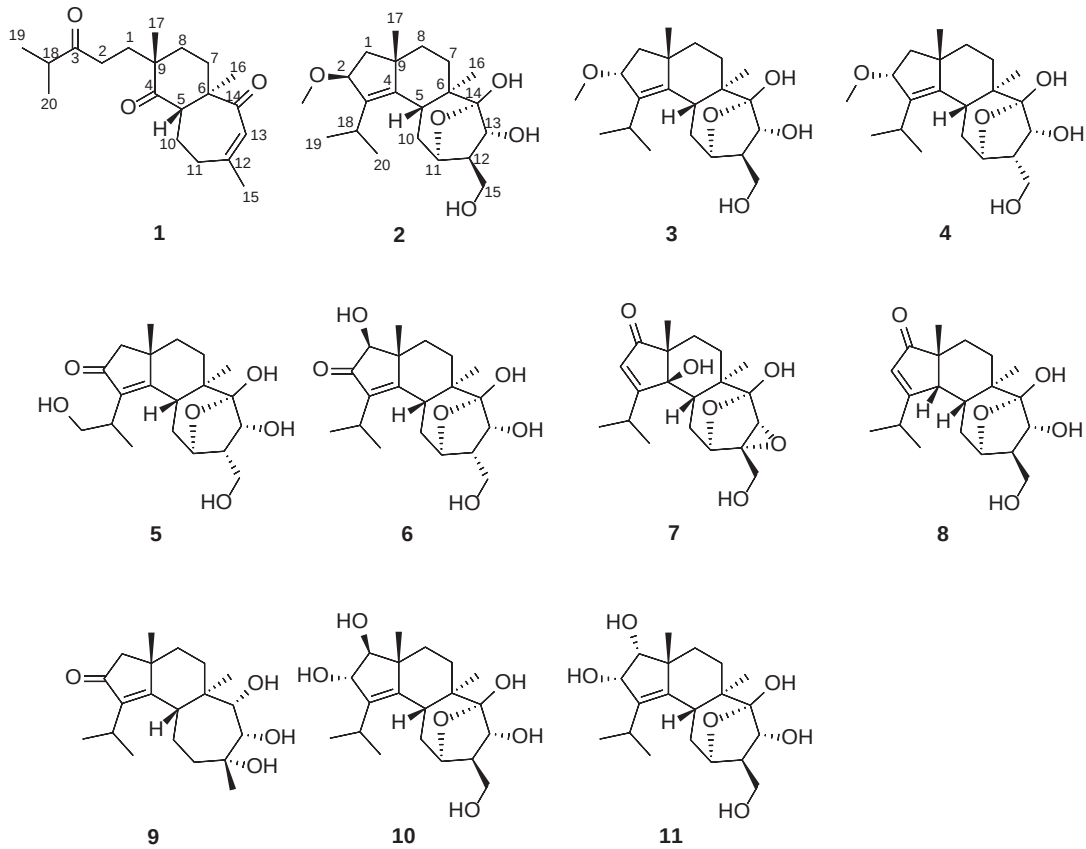

Fig. 1. Structures of compounds 1–11.

2. Results and discussion

2.1. Chemistry

Compound **1** was isolated as white powder. The molecular formula of **1** was determined to be $C_{20}H_{30}O_3$ on the basis of analysis of the NMR data and HRESIMS at *m/z* 319.2271 $[M + H]^+$, indicating six degrees of unsaturation. The IR strong absorptions at 3431 and 1693 cm^{-1} of **1** indicated the presence of *α*,*β*-unsaturated conjugated ketone functionalities.

The ^1H NMR and ^{13}C NMR data (Tables 1 and 2) revealed 20 carbon resonances, corresponding to five methyls (three singlets and two doublets), six methylenes, three methines (one olefinic), and six quaternary carbons (one olefinic and three carbonyl). From the COSY spectrum, CH_2-1/CH_2-2, CH_2-7/CH_2-8, CH-5/CH_2-10/CH_2-11 and CH_3-19/CH-18/CH_3-20 were established. The key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-1 to C-2, C-3, C-4, C-9, C-17, from H-16 to C-5, C-6, C-7, C-14, from H-5 to C-4, C-11, C-14 and from H-15 to C-11, C-12, C-13 (Fig. 2) allowed the structural establishment of **1**. Hence, compound **1** was elucidated as analogue of neocyathin M [8], shares 3,4-*seco*-cyathane skeleton.

The relative configuration of **1** was determined through a NOESY experiment (Fig. 2). The key cross-peaks of H-5/17-Me, 17-Me/H-7β, H-5/H-7β and H-8α/16Me in the NOESY spectrum indicated the relative configuration of **1** as shown. We propose the trivial name cyafricanin A for the new compound **1**.

Compound **2** was isolated as white powder. The molecular formula of **2** was determined to be $C_{21}H_{34}O_5$, with five degrees of unsaturation, based on the HRESIMS at *m/z* 389.2309 $[M + Na]^+$. The ^1H NMR and ^{13}C NMR data (Tables 1 and 2) revealed 21 carbon resonances, corresponding to five methyls (one oxygenated, two singlets and two doublets), five methylenes (one oxygenated), six methines (three oxygenated), and five quaternary carbons (two olefinic and a ketal carbon). From the COSY spectrum, CH_2-1/CH-2, CH_2-7/CH_2-8, CH-5/CH_2-10/CH-11/CH-12, CH-13/CH-12/CH_2-15 and CH_3-19/CH-18/CH_3-20 were established. The key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-1 to C-2, C-3, C-4, C-9, C-17, from H-2 to C-21, from H-16 to C-5, C-6, C-7, C-14, from H-11 to C-14 and from H-15 to C-11, C-12, C-13 (Fig. 2) allowed the structural establishment of **2**. Hence, the planar structure of **2** was elucidated as the same as that of neocyathin P [8].

The absolute configuration of neocyathin P had been determined to be 2*S*, 5*R*, 6*R*, 9*S*, 11*R*, 12*S*, 13*R*, 14*S* from electronic circular dichroism data. Comparing the ^{13}C NMR data of compound **2** with those of neocyathin P, a conspicuous difference in the chemical shift of C-12 was observed. According to the literature and our previous results, the *α*-oriented C-12 of the cyathanes has a δ_C12 value in the range of 48.9–50.3 ppm, while te *β*-oriented C-12 has a δ_C12 value around 56 ppm [4,14,15]. Accordingly, the chemical shift of C-12 (δ_C 55.9 ppm) in compound **2** indicated that C-12 is *β*-orientated. Except for C-12, the other parts of compound **2** were the same as those of neocyathin P. The conclusion was further determined through a NOESY experiment. The key cross-peak of H-1β/17-Me, H-1β/OMe, H-5/H-11 and H-5/H-15 in the NOESY spectrum indicated the relative configuration of **2** as shown in Fig. 1. We propose the trivial name cyafricanin B for the new compound **2.**

Compound **3** was isolated as white powder. The molecular formula of **3** was determined to be $C_{21}H_{34}O_5$, with five degrees of unsaturation, based on the HRESIMS data at *m/z* 389.2302 $[M + Na]^+$. The ^1H and ^{13}C NMR spectral data (Tables 1 and 2) revealed that compound **3** is an epimer of **2**. Comparing the ^1H NMR data of compound **3** with those of **2**, a conspicuous difference was the chemical shift of H-2 together with its proton spin-spin coupling constants. $\delta_{H\text{-}2}$ of compound **2** (and of neocythin P) was '4.23 ppm, br. d' with *J* coupling constant value of

Table 1
^{1}H (500 MHz) NMR data for compounds 1–6 in methanol-d_4 (J in Hz).

No.	1	2	3	4	5	6
1	1.65, m, 2H	1.84, d (13.7);	2.11, m	2.10, dd (12.8, 6.7)	2.20, s, 2H	3.59, s
		1.58, m	1.40, dd (12.8, 6.1)	1.39, dd (12.8, 6.1)		
2	2.59, m; 2.52 m	4.23, br. d (6.6)	4.45, td (6.7, 2.2)	4.46, td (6.7, 2.2)		
5	3.17, d (8.2)	2.35, dd (13.3, 3.7)	2.35, m	2.42, m	2.78, dd (12.7, 3.5)	2.79, dd (12.8, 3.8)
7	2.10, td (13.4, 4.3)	1.54, m;	1.55, m	1.59, m;	1.73, m	1.77, m
	1.71 m	1.45, m	1.46, m	1.29 m	1.58, m	1.59, m
8	1.87, td (14.3, 4.8)	1.62, m;	1.63, m	1.64, m, 2H	1.81, m	1.64, m
	1.70, m	1.45, m	1.61, m		1.70, m	1.60, m
10	2.32, m	2.25, td (13.2, 3.7);	2.18, m	2.18; td (12.8, 3.6) ;	2.50, td (12.8, 3.2)	2.38, m
	1.56, m	1.71, m	1.72, m	1.49; m	1.51, m	1.54, m
11	2.44, m ; 2.36, m	4.32, m	4.29, m	4.17; m	4.23, m	4.23, m
12		2.44, m	2.44, m	2.27; m	2.34, m	2.33, m
13	5.88, s	4.02, d (4.8)	4.01, d (4.9)	4.42; d (7.8)	4.48, d (7.7)	4.49, d (7.8)
15	1.94, s	3.88, dd (10.9, 6.5)	3.87, dd (11.0, 6.5);	3.81; dd (10.9, 5.9);	3.83, m	3.84, dd (10.9, 5.80)
		3.65, dd (10.9, 10.9)	3.62, dd (11.0, 10.0)	3.57; dd (10.9, 8.8)	3.59, m	3.61, m
16	0.98, s	1.01, s	1.11, s	1.10, s	1.09, s	1.03, s
17	1.22, s	1.18, s	1.05, s	1.07, s	1.28, s	1.32, s
18	2.69, m	3.02, m	2.95, m	2.92, m	3.08, m	3.01, m
19	1.09, d (6.9)	1.06, d (6.8)	1.14, d (7.0)	1.14, d (6.8)	3.84, m; 3.65, m	1.16, d (7.0)
					3.68 m	
20	1.09, d (6.9)	1.04, d (6.8)	1.02, d (7.0)	1.01, d (6.8)	1.10, d (6.7)	1.24, d (7.0)
OCH_3		3.25, s	3.25, s	3.25, s		

6.6 Hz, while δ_H-2 of compound **3** was '4.45 ppm, td' with *J* coupling constant value of 6.7, 2.2 Hz [15,16]. The above difference suggested that OMe at C-2 was *α*-oriented in compound **3**. The conclusion was further confirmed by the NOESY correlation of H-2 and Me-17. Careful analysis of the 2D NMR data indicated that other parts of compound **3** were exactly the same as **2**. Finally, the structure with relative configuration of compound **3** was elucidated as shown, and was named as cyafricanin C.

Compound **4** was isolated as white powder. The molecular formula of **4** was determined to be $C_{21}H_{34}O_5$, with five degrees of unsaturation, based on the HRESIMS data at *m/z* 389.2307 [M + Na]$^+$. The ^{1}H NMR and ^{13}C NMR data suggested that **4** shared the same planar structure with **2** and **3**, which was further supported by the HMBC and COSY spectra. δ_H-2 of compound **4** was '4.46 ppm, td' with a *J* coupling constant value of 6.7, 2.2 Hz (i.e., the same as in compound **3**), suggesting that OMe at C-2 was also *α*-oriented. This conclusion was even confirmed by the NOESY correlation of H-2 and Me-17. The chemical shift of C-12 (δ_C 49.5) in compound **4** indicated that C-12 was *α*-orientated. Further analysis of 2D NMR supported the structure of **4** as shown. We propose the trivial name cyafricanin D.

Compound **5** was isolated as white powder. The molecular formula of **5** was determined to be $C_{20}H_{30}O_6$, with six degrees of unsaturation, based on the HRESIMS data at *m/z* 389.1946 [M + Na]$^+$. The ^{1}H NMR and ^{13}C NMR data (Tables 1 and 2) revealed 20 carbon resonances, corresponding to three methyls (two singlets and one doublets), six methylenes (two oxygenated), five methines (two oxygenated), and six quaternary carbons (two olefinic one carbonyl and a ketal carbon). From the COSY spectrum, CH_2–7/CH_2–8, CH-5/CH_2–10/CH-11, CH-13/CH-12/CH_2–15 and CH_2–19/CH-18/CH_3–20 were established. The key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-1 to C-2, C-3, C-4, C-8, C-9, C-17, from H-16 to C-5, C-6, C-7, C-14, from H-11 to C-14 and from H-15 to C-11, C-12, C-13 (Fig. 2) established the planar structure of **5**.

The relative configuration of **5** was determined through a NOESY experiment (Fig. 2). The key cross-peak of H-5/17-Me, H-5/H-12 and H-5/H-13 in the NOESY spectrum determined that Me-17, H-5 were *β*-oriented and Me-16, C-15 and OH-13 were *α*-oriented. Hence, the structure of **5** was elucidated as shown, and named as cyafricanin E.

Table 2
^{1}H (500 MHz) NMR data for compounds 7–11.

No.	7	8	9	10	11
1			2.16, m, 2H	3.59, d (6.9)	3.59, d (2.9)
2	5.93, s	5.90, s		4.47, dd (6.9, 2.6)	4.47, d (2.9)
4		2.76, dd (5.2, 2.0)			
5	2.27, m	2.01, m	2.69 dd (12.4, 1.9)	2.30, m	2.34, m
7	1.70, m	1.59, m	1.78, m	1.53, m, 2H	1.59, m
	1.45, m	1.32, m	1.58, td (14.4, 4.5)		1.49, m
8	1.66, m	1.77, m	1.81, m	1.66, m	1.81, td (13.5, 4.8)
	1.50, m	1.38, m	1.42, m	1.53, m	1.29, m
10	2.17, m, 2H	2.25, td (13.6, 4.0)	2.20, m	2.09, td (13.2, 4.0)	2.29, overlap
		1.61 m	1.90, m	1.75, dd (13.2, 4.0)	1.66, br d (11.9)
11	4.29, m	4.32, m	2.11, m; 1.36, m	4.28, m	4.29, m
12		2.44 m		2.43, m	2.44, m
13	3.66, s	3.93, d (4.9)	2.78, d (6.3)	4.00, d (4.9)	4.03, d (4.8)
14			3.40, d (6.3)		
15	4.19, d (13.3)	3.90, dd (11.2, 6.4)	1.38, s	3.87, dd (11.0, 6.5)	3.89, dd (11.0, 6.5)
	3.81, d (13.3)	3.72, dd (11.2, 9.3)		3.63, dd (11.0, 9.8)	3.64, dd (11.0, 9.6)
16	1.01, s	1.03, s	0.94, s	1.09, s	1.02, s
17	1.09, s	1.14, s	1.25, s	0.97, s	1.18, s
18	2.95, m	2.94, m	3.15, m	3.00, m	3.05, m
19	1.27, d (6.8)	1.17, d (6.5)	1.20, d (6.9)	1.19, d (7.1)	1.13, d (7.0)
20	1.27, d (6.8)	1.27, d (6.5)	1.22, d (6.9)	1.11, d (7.1)	1.11, d (7.0)

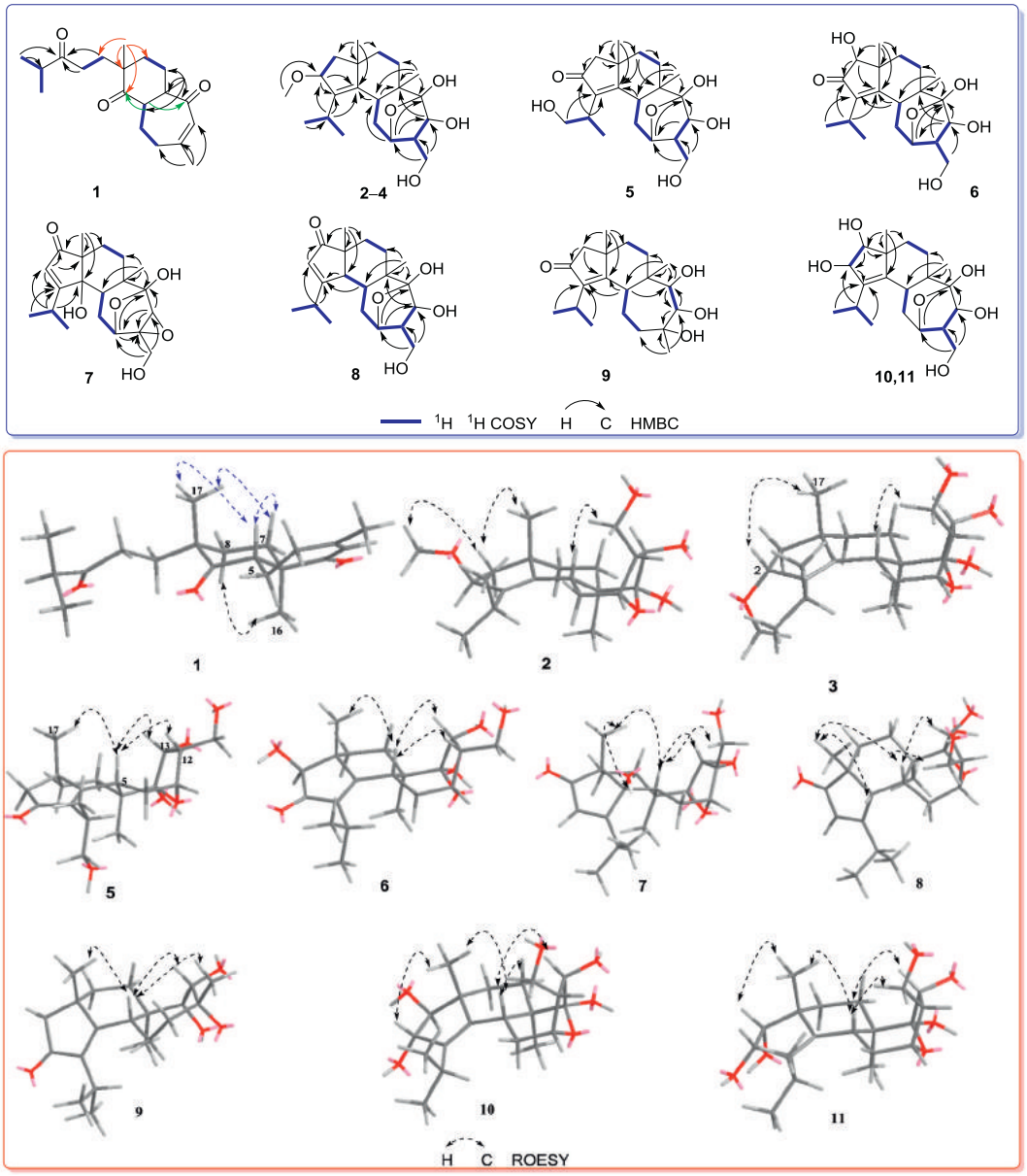

Fig. 2. Selected key 2DNMR correlations of compounds 1–11.

Compound **6** was isolated as yellow oil. The molecular formula of **6** was determined to be $C_{20}H_{30}O_6$, with six degrees of unsaturation, based on the HRESIMS data at *m/z* 389.1951 $[M + Na]^+$. The ^{1}H NMR and ^{13}C NMR data (Tables 1 and 2) of **6** resembled those of Cyathin P [4] except for the chemical shifts of C-1(δ_C 84.5ppm in cyathin P, δ_C 81.1ppm in compound **6**). This difference led to the conclusion that hydroxyl group at C-1 in compound **6** was *β*-oriented, which was supported by the cross-peak of H-1/H-15 in the NOESY spectrum. The other relative configurations in **6** were determined through a NOESY experiment (Fig. 2). The key cross-peak of H-5/17-Me, H-5/H-12 and H-5/H-13 in the NOESY spectrum indicated the structure of 6 as shown and named as cyafricanin F.

Compound **7** was isolated as white powder. The molecular formula of **7** was determined to be $C_{20}H_{28}O_6$, with seven degrees of unsaturation, based on the HRESIMS data at *m/z* 387.1787 $[M + Na]^+$. The ^{1}H NMR and ^{13}C NMR data (Tables 2 and 3) revealed 20 carbon resonances, corresponding to four methyls (two singlets and one doublets), four methylenes (one oxygenated), five methines (one olefinic and two oxygenated), and seven quaternary carbons (two oxygenated, one olefinic, one carbonyl and a ketal carbon). From the COSY spectrum, CH_2-7/CH_2-8, CH-5/CH_2-10/CH-11 and CH_2-19/CH-18/CH_3-20 were established. The key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-2 to C-1, C-3, C-4, C-9, from H-16 to C-5, C-6, C-7, C-14, from H-11 to C-14 and from H-15 to C-11, C-12, C-13 (Fig. 2) determined the planar structure of **7** as shown.

The relative configuration of **7** was determined by key cross-peaks of H-5/17-Me and H-5/H-13 in the NOESY spectrum in CD_3OD. The hydroxyl group at C-4 of **7** was determined in DMSO-d_6 according to cross-peak of 17-Me/4-OH in the NOESY spectrum indicated the relative *β*-configuration for 4-OH (Fig. 2). Thus, all these data allowed us to establish compound **7** and named as cyafricanin G.

Compound **8** was isolated as white powder. The molecular formula of **8** was determined to be $C_{20}H_{30}O_5$, with six degrees of unsaturation, based on the HRESIMS data at *m/z* 349.2023 $[M - H]^-$. The ^{1}H NMR and ^{13}C NMR data (Tables 2 and 3) of **8** resembled those of neocyathin I [7] except for the chemical shifts of C-12. The chemical shift of C-12 (δ_C 56.0 ppm) determined the relative configuration to be *β*-orientated [4]. The relative configuration of **8** was determined by the key cross-peaks

Table 3
^{13}C (125 MHz) NMR data for compounds 1–11.

No.	1	2	3	4	5	6	7	8	9	10	11
1	34.3	46.0	47.5	47.5	53.1	81.1	213.8	215.7	52.8	90.4	86.9
2	36.4	86.3	87.7	87.7	211.1	209.3	127.3	126.1	211.2	83.3	83.8
3	217.4	141.5	141.4	141.4	141.7	142.7	189.0	192.7	144.5	140.9	140.8
4	215.6	144.2	142.1	142.2	180.0	174.3	83.2	53.0	180.0	138.6	140.7
5	51.9	39.0	38.8	38.6	40.6	40.6	44.9	37.7	51.4	39.6	39.0
6	56.7	45.0	44.6	44.4	46.0	47.1	42.7	42.2	48.0	43.9	45.5
7	33.6	29.7	29.9	30.4	29.7	28.9	29.3	29.6	33.3	29.5	29.0
8	33.3	38.6	38.0	38.1	36.8	31.1	31.7	30.6	38.5	36.0	29.1
9	47.3	48.1	46.9	47.0	43.3	46.2	55.0	50.2	43.6	46.9	49.8
10	21.3	27.5	27.6	32.1	31.6	31.3	25.7	29.1	27.0	28.1	27.5
11	37.3	74.5	74.4	76.5	76.3	76.4	74.3	74.7	36.3	74.3	74.6
12	157.7	55.9	55.9	49.5	49.2	49.4	64.1	56.0	61.1	55.9	56.0
13	127.5	71.8	71.7	70.0	70.2	70.2	56.9	71.8	67.7	71.7	72.0
14	208.4	106.5	106.4	107.5	107.4	107.5	104.6	106.9	81.1	106.2	106.6
15	27.0	60.6	60.7	62.1	62.0	62.0	58.9	60.5	22.5	60.7	60.7
16	17.6	13.2	13.3	13.0	13.3	13.4	14.9	15.9	11.8	13.1	13.1
17	24.1	26.7	26.7	26.7	26.5	24.8	16.2	21.4	23.6	19.4	25.7
18	41.9	27.2	27.2	27.2	34.9	26.1	31.6	33.1	27.1	27.2	27.4
19	18.6	23.1	24.7	24.6	65.3	20.2	24.7	23.3	20.8	24.7	24.0
20	18.6	23.1	20.0	20.0	15.5	21.0	22.1	21.0	20.2	19.4	21.3
OCH_3		56.0	55.9	55.9							

of H-5/17-Me, H-5/H-13 and H-4/17-Me in the NOESY spectrum. We propose the trivial name cyafricanin H for the new compound **8**.

Compound **9** was isolated as white powder. The molecuar formula of **9** was determined to be $C_{20}H_{32}O_4$, with five degrees of unsaturation, based on the HRESIMS data at *m/z* 335.2275 $[M - H]^-$. The ^1H NMR and ^{13}C NMR data (Tables 2 and 3) revealed 20 carbon resonances, corresponding to five methyls (three singlets and two doublets), five methylenes, four methines (two oxygenated), and six quaternary carbons (one oxygenated, two olefinic and one carbonyl). Further analysis of the 1D NMR data together with the HMBC and COSY spectra allowed the structural establishment of **9**. From the COSY spectrum, CH_2–7/CH_2–8, CH-5/CH_2–10/CH_2–11, CH-13/CH-14 and CH_2–19/CH-18/CH_3–20 were established. The key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-18 to C-2, C-3, C-4, from H-16 to C-5, C-6, C-7, C-14, from H-5 to C-3, C-4, C-6, C-9, C-11, C-14 and from H-15 to C-11, C-12, C-13 (Fig. 2) suggested the structure of **9** as shown. The relative configuration of **9** was determined through a NOESY experiment. The key cross-peaks of H-5/17-Me, H-5/H-13, H-5/H-14 and H-5/H-15 in the NOESY spectrum (Fig. 2) established **9** as cyafricanin I.

Compound **10** was isolated as white powder. The molecular formula of **10** was determined to be $C_{20}H_{32}O_6$, with five degrees of unsaturation, based on the HRESIMS data at *m/z* 391.2086 $[M + Na]^+$. The ^1H NMR and ^{13}C NMR data (Tables 2 and 3) revealed 20 carbon resonances, corresponding to four methyls (two singlets and two doublets), four methylenes (one oxygenated), seven methines (four oxygenated), and five quaternary carbons (two olefinic and a ketal carbon). The $^1H-^1H$ COSY cross-peaks of CH_2-7/CH_2-8, CH-5/CH_2-10, CH-11/CH-12, CH-13/CH-12/CH_2-15, CH-1/CH-2 and CH_2-19/CH-18/CH_3-20 and the key HMBC correlations from H-17 to C-1, C-4, C-8, C-9, from H-1 to C-2, C-8, C-9, C-17, from H-16 to C-5, C-6, C-7, C-14 and from H-15 to C-11, C-12, C-13 established the planar structure of **10** as shown (Fig. 2).

The relative configuration of **10** was determined by H−H coupling constant and the NOESY experiment. The larger coupling constant (J = 6.9 Hz) between H-1 and H-2 suggested the *trans* configuration. The cross-peak of H-2/Me-17 suggested that OH-2 is α-oriented, while OH-1 is β-oriented. On the other hand, the key cross-peaks of H-5/17-Me, H-5/H-15, H-5/H-13, H-8β/17-Me, suggested OH-13 to be α-oriented and C-15 to be β-oriented (Fig. 2). We propose the trivial name cyafricanin J for the new compound **10**.

Compound **11** was isolated as a white powder. The molecular formula of **11** was determined to be $C_{20}H_{32}O_6$, with five degrees of unsaturation, based on the HRESIMS at *m/z* 759.4297 $[2M + Na]^+$. The ^1H NMR and ^{13}C NMR data (Tables 2 and 3) of **11** resembled those of compound **10** except for the configuration of C-1 based on the chemical shift of C-1 (90.4ppm in **10**; 86.9ppm in **11**). The smaller coupling constant value of 2.9 Hz between H-1 and H-2 indicated that OH-1 and OH-2 are *cis*-oriented in the double bind. In the ROESY spectrum, cross-peaks of H-1/17-Me suggested OH-1 to be α-oriented, which decided OH-2 to be α-oriented, too. The other configurations of **11** were determined by key cross-peaks of H-5/17-Me, H-5/H-11, H-5/H-13 and H-5/H-15. Therefore, the structure of **11** was determined as cyafricanin K.

2.2. Bioactivity

2.2.1. Neurotrophic activity

Compounds **1–11** were evaluated for their neuritogenic activity using rat pheochromocytoma (PC-12) cells as a model system of neuronal differentiation, and NGF (20 ng/mL) was used as positive control. Compounds **1–11** in combination with NGF (20 ng/mL) were found to exert a significant increase in neurite-bearing cells compared to NGF-treated PC-12 cells at concentrations of 20 μM (Fig. 3).

2.2.2. Anti-inflammatory activities

We firstly examined the cytotoxicity of the isolated compounds in BV2 microglial cells. As shown in Fig. 5, all compounds at the concentrations from 5 μM to 100 μM for 24 h did not show any toxicity. COX-2 and iNOS enzymes are two major inflammatory modulators in brain neurodegeneration. Subsequently, we examined the effects of all of compounds on the expression of iNOS and COX-2 in LPS-stimulated BV2 microglial cells. Based on Western blot analysis as shown in Fig. 4, compound **11** could significantly suppress LPS-induced COX-2 expression, while compound **2** could markedly inhibit LPS-induced iNOS expression. Particularly, compound **1** showed strong inhibitory effects on both COX-2 and iNOS expression. In addition, nitric oxide (NO) from L-arginine oxidized after stimulation is the product of the enzyme iNOS. Actually, compounds **1** and **2** significantly suppressed LPS-induced NO production in culture medium (Fig. 4).

2.2.3. Molecular docking simulations of the binding of compound 1 to iNOS

According to our previous studies, the bioactive diterpenoids normally showed stronger interactions with iNOS protein than COX-2. Consistently, we elucidated the inhibitory effects of compound **1** on

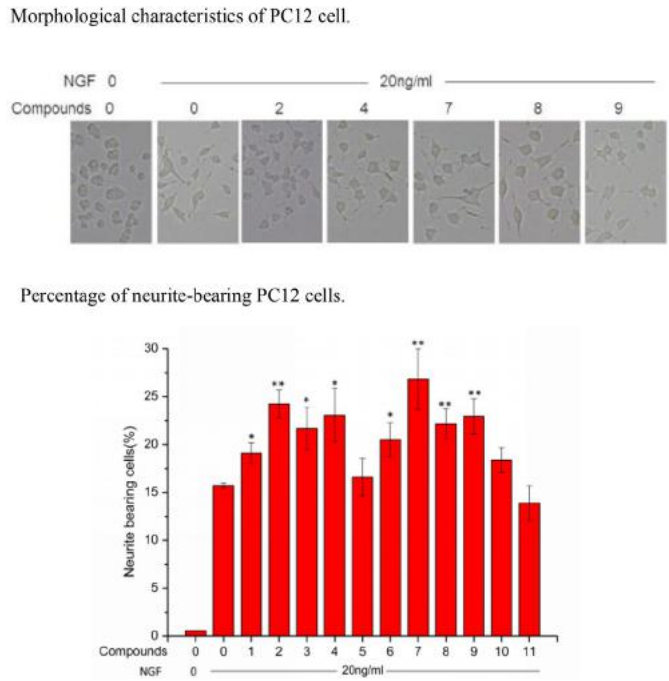

Fig. 3. Effects of compounds 1–11 on the NGF-promoted neurite outgrowth in PC-12 cells.
PC-12 cellswere seeded into poly-L-lysine-coated 24-well plates in normal serum medium for 24 h until adherence, shifted to low serum medium (2% HS and 1% FBS). PC-12 cells were exposed to 0.1% DMSO and 20 ng/mL NGF as a negative control and 50 ng/mL NGF as a positive control, respectively. Cell morphology was observed and photographed as described in Measurement of neurite outgrowth. Neurite bearing cells were analyzed as described in Measurement of neurite outgrowth. Data represent the mean ± SD of three independent experiments. $^{*}p < .05$ and $^{**}p < .01$ represent significant differences compared with NGF-treated PC-12 cells (positive control) (ANOVA followed by Dunnett's posthoc multiple comparisons test).

iNOS protein using molecular docking. Molecular docking study indicated strong interactions between compound **1** and iNOS protein. The logarithm of free binding energy was −9.13 kcal/mol. Compound **1,** with strong NO inhibitory activity, matched well in the protein-binding pocket by hydrogen bonding with TYR373, and there is also week electrostatic interaction between oxygen of the carbanyl group at C-3 and the iron atom in the ferriporphyrin (Fig. 5).

3. Materials and methods

3.1. General experimental procedures

Optical rotations were recorded on an Autopol III automatic polarimeter (Rudolph Research Analytical). UV and IR spectra were obtained on a Thermo Scientific Evolution-300 UV − visible spectrophotometer and a Bruker Tensor 27 FT-IR spectrometer with KBr pellets. NMR spectra were obtained on Bruker Avance 800 MHz and Bruker Avance III 500 spectrometers with tetramethylsilane as an internal standard at room temperature. High-resolution (HR) ESIMS spectra were recorded on a Thermo Fisher Scientific Q-TOF mass spectrometer. Silica gel (300–400 mesh, Qingdao Marine Chemical Ltd., People's Republic of China) and RP- 18 gel (20–45 μm, Fuji Silysia Chemical Ltd., Japan) were used for column chromatography (CC). Semipreparative HPLC was performed on a Waters 1525EF liquid chromatography system equipped with a Hypersil BDS C18 column (4.6 mm × 250 mm; 10.0 mm × 250 mm) and a Chiral column Chiralpak AD-H (4.6 mm × 250 mm). Fractions were monitored by TLC. The compounds were visualized by heating silica gel plates immersed in 10% H_2SO_4 in ethanol.

3.2. Fungal material

The fungus *Cyathus africanus* was purchased from China General Microbiological Culture Collection Center (CGMCC) with the accession number CGMCC 5.1117. A specimen (No. CA20150823) was deposited at the College of Chemistry & Pharmacy, Northwest A&F University, Shaanxi, China. The fermentation was carried out in one hundred 500 mL Fernbach flasks each containing 80 g of rice and 120 mL of distilled water. All of the flasks were incubated at 28 °C for 180 days.

3.3. Extraction and isolation

The fermented rice sample was extracted thoroughly with methanol, and the solutions were filtered and evaporated under reduced pressure. The methanol extract of fermented rice was diluted with water to 5 L, then extracted with ethyl acetate (5 L × 3). The EtOAc layer was concentrated under reduced pressure to give a crude extract (43.0 g), which was applied to a RP-18 column eluting with a gradient of MeOH − H_2O (10%–100%) to obtain five fractions, A − F. Fraction E (509 mg) was further fractionated by silica gel CC eluting with a gradient of $CHCl_3$ − MeOH (50:1–10:1) to yield five fractions (E1–E5). Fraction E2 was subjected to Sephadex LH-20 (MeOH) and further purified by semipreparative HPLC (15%, MeOH − H_2O, 2 mL/min) to yield **5** (t_R = 46.0 min, 12.9 mg). Fraction E3 was subjected to Sephadex LH-20 (MeOH) and further purified by semipreparative HPLC (30%, MeOH − H_2O, 2 mL/min) to yield **6** (t_R = 37.0 min, 5.0 mg) and **7** (t_R = 27.0 min, 8.1 mg). Fraction D (1 .0g) was further purified by silica gel CC eluting with a gradient of $CHCl_3$ − MeOH (50:1–10:1), to yield five fractions (D1–D5). Fraction D3 was subjected to Sephadex LH-20 (MeOH) and silica gel CC ($CHCl_3$ − MeOH, 20:1) and further purified by semipreparative HPLC (30%, MeOH − H_2O, 2 mL/min) to yield **1** (t_R = 29.0 min, 1.6 mg). Fraction D4 was subjected to Sephadex LH-20 (MeOH), and further purified by semipreparative HPLC (60%, MeOH − H_2O, 2 mL/min) to yield **3** (t_R = 34.0 min, 2.9 mg). Fraction D5 was subjected to Sephadex LH-20 (MeOH) and silica gel CC ($CHCl_3$ − MeOH, 50:1) and further purified by semipreparative HPLC (65%, MeOH − H_2O, 2 mL/min) to yield **2** (t_R = 20.0 min, 4.9 mg) and **4** (t_R = 23.0 min, 4.2 mg). Fraction C (2.7 g) was further purified by silica gel CC eluting with a gradient of $CHCl_3$ − MeOH (50:1–10:1), to yield five fractions (C1–C5). Fraction C4 was purified by semipreparative HPLC (35%, MeOH − H_2O, 2 mL/min) to yield **8** (t_R = 42.0 min, 4.0 mg). Fraction B (3.3 g) was further purified by silica gel CC eluting with a gradient of $CHCl_3$ − MeOH (50:1–10:1), to yield five fractions (B1–B5). Fraction B4 was purified by semipreparative HPLC (55%, MeOH − H_2O, 2 mL/min) to yield **9** (t_R = 35.0 min, 2.0 mg). Fraction F (152 mg) was further purified by silica gel CC eluting with a gradient of $CHCl_3$ − MeOH (50:1–10:1), to yield five fractions (F1–F5). Fraction F2 was purified by semipreparative HPLC (20%, hexane − isopropanol, 1 mL/min) to yield **11** (t_R = 20.0 min, 20.0 mg). Fraction F3 was subjected to Sephadex LH-20 (MeOH) and silica gel CC ($CHCl_3$ − MeOH, 40:1) and further purified by semipreparative HPLC (20%, hexane − isopropanol, 1 mL/min) to yield **10** (t_R = 40.0 min, 10.0 mg). Fraction F4 was purified by semipreparative HPLC (50%, MeOH − H_2O, 2 mL/min) to yield **12** (t_R = 15.0 min, 1.0 mg).

Compound ***(1):*** white powder; $[\alpha]_D^{25}$–8.2 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 204 (3.23), 239 (3.36) nm; IR (KBr) ν_{max} 3422, 2954, 2839, 2121, 1646, 1460, 1408, 1018, 689 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 319.2271 $[M + H]^+$ (calcd. for $C_{20}H_{30}O_3H$, 319.2273).
Compound ***(2):*** white powder; $[\alpha]_D^{25}$ + 33 (*c* 0.04, MeOH); UV (MeOH) λ_{max} (log ε) 213 (3.25) nm; IR (KBr) ν_{max} 3442, 2965, 2383, 2311, 1640, 1049, 1017, 683 cm^{-1}; ^1H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 389.2309 $[M + Na]^+$ (calcd. for $C_{21}H_{34}O_5Na$, 389.2304).
Compound ***(3):*** white powder; $[\alpha]_D^{25}$–18.0 (*c* 0.04, MeOH); UV

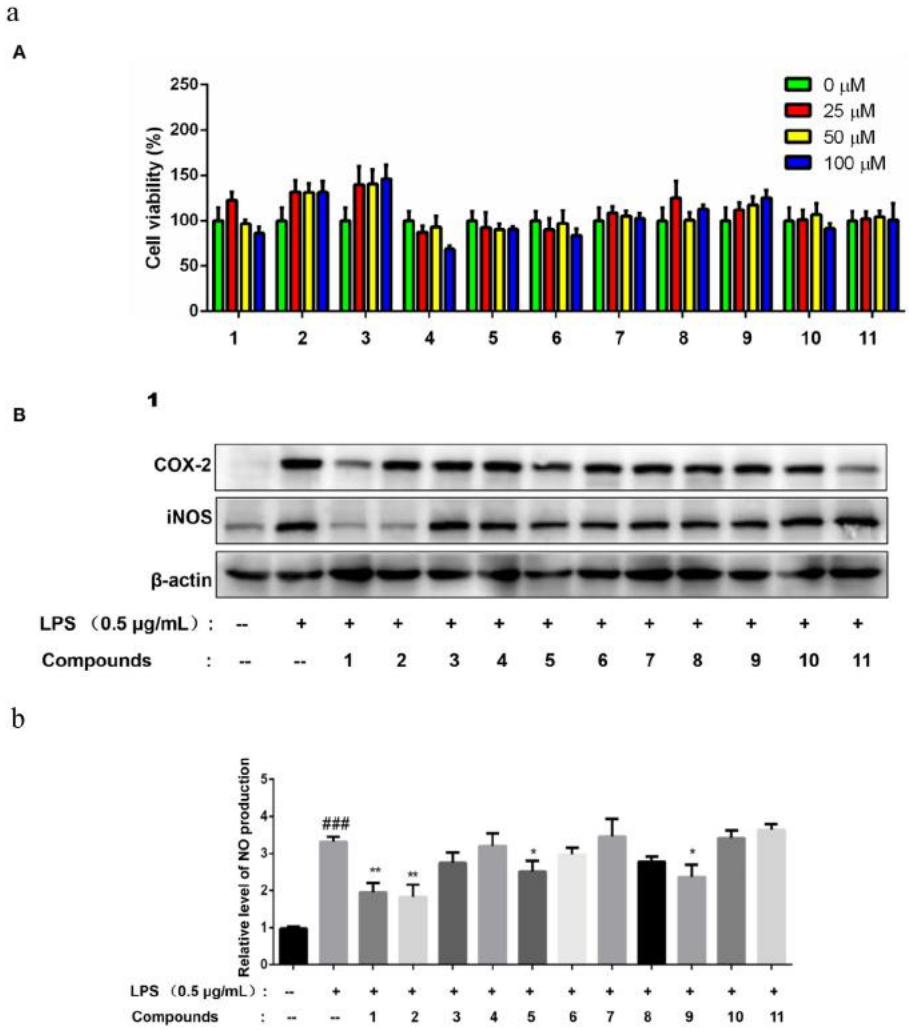

Fig. 4. (a) Effects of isolated compounds on LPS-induced iNOS and COX-2 expression in BV2 microglia cells.(A) The toxicity of isolated compounds in BV2 microglia cells. The cells were treated with all compounds at the concentration of 0, 25, 50 and 100 μM for 24 h, and the cell viability was determined by MTT assay. (B) Analysis of iNOS and COX-2 expression by Western blot. The cells were pre-treated with all compounds with the concentration of 50 μM for 1 h, and subsequently incubated with 0.5 μg/mL of LPS for 24 h. The expression of iNOS and COX-2 was examined by Western blot. Chemical shifts and proton spin-spin coupling constants of H-2 in related compounds. (b) Effects of isolated compounds on NO production in LPS-stimulated BV2 microglia cells. BV2 cells were pre-treated with all compounds (50 μM) for 1 h, and stimulated with 0.5 μg/mL LPS for 24 h. After the drug treatment as previously described, he culture medium was collected for measurement of NO production by using Griess reagent kit for nitrite determination. Data were expressed as mean ± SEM ($n = 3$) and analyzed by one-way ANOVA with Fisher's LSD test. *p < .05; **p < .01; (Drug plus LPS vs LPS).

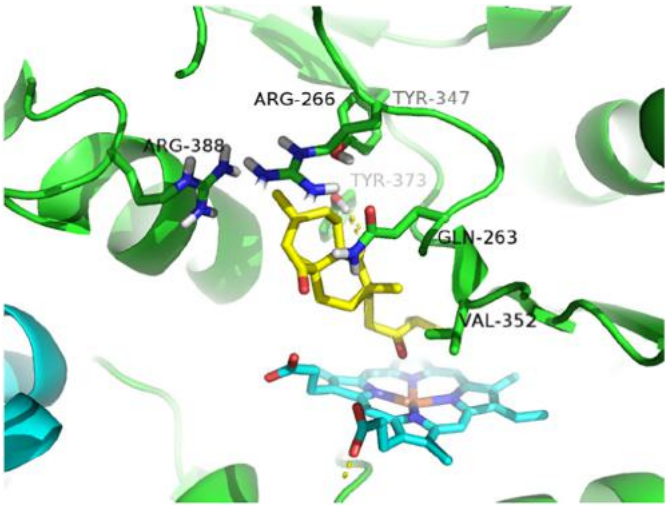

Fig. 5. Molecular docking simulations of the binding of compound 1 to iNOS. Virtualdrug-protein complexes including compound **1** for iNOS was obtained at lowest energy conformation, while potential hydrogen contacts between compound and protein were highlighted by dashed lines. Atoms were colored as follows: carbon, yellow; nitrogen, blue; oxygen, red; hydrogen, gray.

(MeOH) λ_{max} (log ε) 213 (3.18) nm; IR (KBr) ν_{max} 3388, 2950, 2835, 2524, 2040, 1653, 1459, 1021, 680 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 389.2302 $[M + Na]^+$ (calcd. for $C_{21}H_{34}O_5Na$, 389.2304).

Compound ***(4)***: white powder; $[\alpha]_D^{25}$ - 26.0 (*c* 0.02, MeOH); UV (MeOH) λ_{max} (log ε) 209 (3.43) nm; IR (KBr) ν_{max} 3451, 2383, 2336, 1638, 1020, 646 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 389.2307 $[M + Na]^+$ (calcd. for $C_{21}H_{34}O_5Na$, 389.2304).

Compound ***(5)***: white powder; $[\alpha]_D^{25}$–24.5 (*c* 0.06, MeOH); UV (MeOH) λ_{max} (log ε) 196 (3.92), 243 (4.22) nm; IR (KBr) ν_{max} 3394, 2950, 2836, 2333, 2040, 1653, 1459, 1021, 681 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 389.1946 $[M + Na]^+$ (calcd. for $C_{20}H_{30}O_6Na$, 389.1940).

Compound ***(6)***:yellow oil; $[\alpha]_D^{25}$–22.5 (*c* 0.05, MeOH); UV (MeOH) λ_{max} (log ε) 192 (3.78), 244 (4.08) nm; IR (KBr) ν_{max} 3451, 2080, 1638, 1462, 1387, 1017, 678 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 1 and 3; positive HRESIMS *m/z* 389.1951 $[M + Na]^+$ (calcd. for $C_{20}H_{30}O_6Na$, 389.1940).

Compound ***(7)***: white powder; $[\alpha]_D^{25}$–25.3 (*c* 0.08, MeOH); UV (MeOH) λ_{max} (log ε) 199 (3.86), 230 (4.02) nm; IR (KBr) ν_{max} 3419, 2954, 2840, 2521, 2080, 1646, 1461, 1018, 682 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 2 and 3; positive HRESIMS *m/z* 387.1787 $[M + Na]^+$ (calcd. for $C_{20}H_{28}O_6Na$, 387.1784).

Compound ***(8)***: white powder; $[\alpha]_D^{25}$–32.5 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 200 (3.71), 236 (3.89) nm; IR (KBr) ν_{max} 3448, 2962, 2382, 2082, 1641, 1404, 1109, 1017, 681 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 2 and 3; negative HRESIMS *m/z* 349.2023 $[M - H]^-$ (calcd. for $C_{20}H_{29}O_5$, 349.2015).

Compound ***(9)***: white powder; $[\alpha]_D^{25}$–37.1 (*c* 0.2, MeOH); UV

(MeOH) λ_{max} (log ε) 195 (3.50), 243 (3.84) nm; IR (KBr) ν_{max} 3417, 2953, 2838, 2521, 2123, 1648, 1460, 1019, 684 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 2 and 3; negative HRESIMS *m/z* 335.2275 $[M - H]^-$ (calcd. for $C_{20}H_{31}O_4$, 335.2222).

Compound ***(10)***: white powder; $[\alpha]_D^{25}$–4.5 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 210 (3.60) nm; IR (KBr) ν_{max} 3389, 2949, 2835, 2524, 2175, 1653, 1457, 1023, 677 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 2 and 3; positive HRESIMS *m/z* 391.2086 $[M + Na]^+$ (calcd. for $C_{20}H_{32}O_6Na$, 391.2097).

Compound ***(11)***: white powder; $[\alpha]_D^{25}$–19.0 (*c* 0.08, MeOH); UV (MeOH) λ_{max} (log ε) 210 (3.69) nm; IR (KBr) ν_{max} 3446, 2966, 2078, 1638, 1378, 1048, 1018, 609 cm^{-1}; ^{1}H and ^{13}C NMR data, see Tables 2 and 3; positive HRESIMS *m/z* 759.4297 $[2M + Na]^+$ (calcd. for $C_{40}H_{64}O_{12}Na$, 759.4295).

3.4. Analysis of neurite outgrowth of PC-12 cells

Morphological analysis and quantification of neurite bearing cells were carried out using phase-contrast microscope as described previously. PC-12 cells were seeded on poly-L-lysine-coated 24-well plates at a density of 2×10^4 cells/mL in normal serum medium for 24 h. The F-12 medium containing low serum (1% HS and 0.5%FBS) was replaced prior to exposure to vehicle (0.1% DMSO) or indicated reagents. The cells were treated with tested compounds at concentration of 10 μM with 20 ng/mL of NGF. Cells without treatment served as a negative control. Cells treated with 20 ng/mL of NGF served as a positive control. One concentration experiment was repeated in three wells. After an additional 48 h of incubation, neurite outgrowth of PC-12 cells was observed under an inverted microscope using phase-contrast objectives and photographed by the digital camera. Five images were selected randomly under a microscope for each well. At least 100 cells in each of five randomly separated fields were scored and the proportion of cells with neuritis greater than or equal to the length of one cell body were positive for neurite outgrowth, and expressed as a percentage of the total cell number in five fields. Experiments were repeated at least three times and data are expressed as mean ± SD.

3.5. Cell culture and drug treatment

BV2 microglial cells were maintained in Dulbecco's Modified Eagle Medium (DMEM) containing 10% fetal bovine serum (FBS) and 1% penicillin/streptomycin (Invitrogen, Carlsbad, CA, USA) at 37 °C in a humidified 5% CO_2 atmosphere. For drug treatment, the cells at 70–80% confluence in DMEM with 3% FBS were treated with isolated compounds at 50 μM for the specified times, whereas the control cells were treated with equal amount of dimethyl sulfoxide (DMSO) under the same conditions. For Western blot analysis, the cells were pretreated with compounds for 1 h and subsequently stimulated with LPS for another 24 h.

3.6. Cell viability assay

Cell viability was measured by MTT assay. Briefly, BV2 microglia cells were seeded in 96 well plates overnight and treated with different concentrations of isolated compounds for 24 h or 72 h. After incubation and removal of culture medium, the BV2 cells were incubated with 0.5 mg/mL MTT in phosphate-buffered saline (PBS) for 4 h. Then the MTT was removed and resolved with 150 μL of DMSO. The production of purple formazan was detected for the absorbance at 570 nm on a Bio-Rad microplate reader (Hercules, CA, USA). The cell viability was presented as a percentage to that of controls.

3.7. Western blot analysis

After drug treatment, the cellular proteins were extracted and analyzed via Western blotting analysis for protein expression as previously described [7,17]. Briefly, the cellular proteins were isolated and subjected to electrophoresis on 10% SDS-polyacrylamide gels and transferred onto a polyvinylidenedifluoride (PVDF) membrane. After overnight incubation in 5% BSA in the mixture of Tris-buffered saline and Tween-20 (TBS-T), the membranes were probed with primary antibodies: iNOS, COX-2 and GAPDH antibodies (Cell Signaling Technology, Boston, MA, USA) for 4 h and subsequently detected by goat anti-rabbit IgG-HRP conjugate secondary antibodies. The activity of peroxidase retained on the blots was measured by enhanced chemiluminescence (ECL) detection reagents from GE Healthcare (Uppsala, Sweden) according to the manufacturer's instruction.

3.8. Measurement of NO production in culture medium

After drug treatment, nitrite in culture medium was measured to reflect NO production by using Griess reagent kit for nitrite determination (ThermoFisher Scientific, Waltham, MA, USA) [18]. In brief, 200 μL culture medium, 80 μL milli-Q water and 20 μL Griess reagent were mixed and added into 96 well plate. After incubation at room temperature for 30 min, the absorbance was determined by a spectrophotometric microplate reader at 540 nm.

3.9. Molecular docking studies

Molecular docking simulations were performed using the software Autodock4.2 Vina along with AutoDock Tools (ADT 1.5.6) using the hybrid Lamarckian Genetic Algorithm (LGA) as our previous studies [19]. The three dimensional (3D) crystal structure of iNOS (PDB code: 3E7G) was obtained from the RCSB Protein Data Bank. The standard 3D structure (PDB format) of compound **1**was constructed by using the "SKETCH" option function in SYBYL-X, whose configurations were determined by their NOESY spectra and TDDFT ECD calculations. The cubic grid box of 44 Å size (x, y, z) with a spacing of 0.375 Å and grid maps were built. The docking parameters consisted of setting the population size to 150, the number of evaluations to 2,500,000, the number of generations to 270,000, and the number of top individuals that automatically survive to 20, while the number of docking run was set to 40 with other default values during each docking run. The results of the most favorable free energy of binding were chosen as the resultant complex structures.

4. Conclusion

In summary, eleven cyathane-type diterpenoids, cyafricanin A–K (**1–11**) were isolated from solid cultures of the Bird's nest fungus, *Cyathus africanus*. These metabolites were found to exert neurite outgrowth-promoting activity in NGF-mediated PC12 cells. Cyafricanin A (**1**) maybe exhibit anti-neuroinflammatory activity via the inhibition of pro-inflammatory enzymes expression. Furthermore, the molecular docking study revealed that cyafricanin A might express iNOS protein by binding to the active center of iNOS. These metabolites might be useful as lead compounds for development of anti-inflammatory agents and also utilized as neurite outgrowth-promoting agents that could be used in therapy of AD and related neurodegenerative disorders.

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The authors declare no competing financial interests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fitote.2019.02.013.

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